Abin Anil

AI/ML Engineer | Python, GenAI, RAG & Agentic Systems
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SUMMARY

AI/ML Engineer with a strong Python foundation, building GenAI, RAG, and agentic AI applications. Backed by 2+ years of technical support engineering experience, including AI/ML project support.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, JavaScript, FastAPI, REST APIs, WebSockets, AsyncIO, PostgreSQL, Pandas, NumPy

GenAI & Agentic AI: LangChain, LangGraph, RAG, MCP, Gemini Live API, Hugging Face, Prompt Engineering, QLoRA

ML & NLP: Scikit-Learn, XGBoost, TensorFlow, PyTorch, ANN, RNN, Transformers, NLTK, spaCy, STT, Faster-Whisper, VAD

Cloud & Tools: Docker, GCP Cloud Run, Jenkins CI/CD, Git, GitHub

PROJECTS

AI-Powered Technical Interview Assistant Oct 2026
GitHub: github.com/abinanil000datascience/ai-interview-assistant

- Built a real-time voice interviewer (FastAPI, WebSockets, Gemini Live API) with bidirectional audio streaming and a 6-step state-machine prompt that handles interruptions and switches to Tutor Mode.
- Engineered an async GitHub pre-fetch pipeline (`httpx`) injecting a candidate's repos, READMEs and core code into the system prompt at session start: **zero mid-call network calls**.
- Cut local STT (`faster-whisper`) latency from **~60 s to <2 s** using Silero VAD silence filtering, byte-alignment checks and `asyncio.Lock` to stop CPU bottlenecks and hallucination loops.

Nexus Intelligence – Multi-Agent AI Trading System Sep 2026
GitHub: github.com/abinanil000datascience/nexus-ai-trading-engine

- Built a four-agent trading system (LangGraph, MCP, async FastAPI) that fuses RSI/MACD signals (Pandas) with Gemini news sentiment into real-time buy/sell verdicts, with a React monitoring dashboard.
- Isolated agent execution in background threads to remove event-loop blocking: **3–5x throughput** and **p95 latency ~2.4 s to <800 ms** (1,000 concurrent tick simulation).

Healthcare IT Ticket Resolution Engine (ITSM) Aug 2026
GitHub: github.com/abinanil000datascience/Healthcare_ITSM_Agent

- Built a RAG pipeline with hybrid search (BM25 + HNSW in pgvector) on Dockerized FastAPI: **2.6x faster retrieval** than an exact L2 scan (10K tickets) with **>97% context precision/recall** (RAGAS).
- Fine-tuned Llama-3 (8B) with Unsloth and QLoRA (4-bit) for local inference, saving **$50/month** versus GPT-4o (2,500 tickets/month).

AI Career Platform – Smart Resume Matching Aug 2026
GitHub: github.com/abinanil000datascience/ai-career-platform

- Built an ATS-style semantic resume-to-job matching engine (LangGraph, embeddings, PostgreSQL/pgvector) on async FastAPI with Gemini and Groq LLMs, deployed on GCP Cloud Run.
- Added async exponential backoff (Tenacity), lifting concurrent 100-resume batches from 85% to **100% eventual success**.

PROFESSIONAL EXPERIENCE

Technical Support Engineer, Feathersoft Aug 2024 – Present
Distributed DNA diagnostics platform

- Supported AI/ML projects, using Python, Pandas, and LLM APIs to analyze model, data, and API issues.
- Maintained 99.8% uptime and data integrity for a distributed diagnostics platform with automated report workflows.
- Cut production errors by 22% by debugging API and data-ingestion failures (cURL, Postman, Swagger).
- Cut MTTR by 20% with SQL fixes; sped up resolution by 15% via Kibana and Grafana log analysis.
- Led client triage (99% CSAT, 120+ escalations).

PROFESSIONAL TRAINING

Data Science With AI Certification – MetaiLearn Nov 2025 – Jun 2026
Structured training that built my foundation in AI/ML engineering

- Learned ML and deep learning fundamentals, then moved into NLP and Large Language Models (LLMs).
- Built RAG pipelines and OpenAI API integrations, applied in the GenAI and agentic projects above.

CERTIFICATIONS

Fundamentals of AI Agents Using RAG and LangChain – IBM Sep 2026
Covered: RAG, LangChain (loaders, retrievers, chains, agents), prompt templates, FAISS, Hugging Face
Credential: coursera.org/verify/S0HTSWVNLSTW

EDUCATION

B.Tech, Computer Science & Engineering 2020 – 2024
University College of Engineering, Thodupuzha